Chapter 2
Shepherding Autonomous Goal-Focused Swarms in Unknown Environments Using Hilbert Space-Filling Paths

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Abstract A novel technique has been developed for autonomous swarm-based unknown environment scouting. A control method known as swarm shepherding was employed, which replicates the behaviour seen when a sheepdog guides a herd of sheep to an objective location. The guidance of the swarm agents was implemented using low computation cost, force-based behaviours. The exploration task was augmented by introducing swarm member role assignments, including a role which imposes a localised covering area for agents which stray too far from the swarm global centre of mass. The agents then proceeded to follow a Hilbert space-filling curve (HSFC) path within their localised region. The simulation results demonstrated that the inclusion of the HSFC paths improved the efficiency of

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Keywords Shepherding · Swarm robotics · Hilbert space-filling curves · Herding · Robotic exploration

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2.1 Introduction

The emergence of intelligent autonomous robots is opening up exciting new opportunities for humanity. Arduous, monotonous, and dangerous tasks, normally carried out by humans, can be delegated to robotic systems, which have the potential to increase efficiency and reduce human casualties when operating in adverse environments.

The exploration of unknown environments, in particular, often presents unforeseen challenges and inherent risks due to the uncertainties involved [18]. Examples of dangerous human exploratory missions to uncharted territories include Antarctic expeditions [13], the navigation of cave systems [14], and the continual exploration of the world's oceans [15].

While single uninhabited systems are capable of completing complex missions, the introduction of multi-robotic teams can permit an increased level of efficiency to a given task, especially when these tasks are tedious or time-consuming [1]. Moreover, the use of swarm robotic systems allows for a higher degree of flexibility and redundancy when analysing unfamiliar surroundings [20].

This study is a step towards solving a problem that has previously been posed by the authors [9], which aimed to distribute a network of uninhabited surface vessels, with each acting as a sea state estimation sensor, in order to determine the wave properties of complex wave systems. Unknown sea states can pose major risks to the maritime industry as wave-induced loads are capable of damaging marine structures and destabilising ocean platforms [6].

Described in this chapter is a novel method for the exploration of unknown environments (with a focus on aquatic environments with wave systems present) which uses a swarm of agents guided by a shepherd to a number of goal locations. The use of shepherding for environment exploration has seldom been explored before (although shepherding for covering was briefly discussed by Lien et al. [8]). The swarm is tasked with finding the goals within an environment, where no information is assumed about the environment prior to the start of mission. Each member in the swarm only has a localised view of the environment, only possessing memory to respond to their assigned role.

The shepherding guidance technique, then, replicates the behaviour observed by sheepdogs when herding sheep to objective destinations [16]. Additionally, certain agents which become separated from the swarm are forced to follow a Hilbert space-filling curve (HSFC) path within localised regions of the mission environment. This allows the swarm agents to explore the environment for goal locations without the aid of the shepherd, where the use of multiple agents in the classic shepherding case could leave the system overly redundant (few agents could be capable of reaching all goals).

The goal-based exploration model allows the swarm of agents to analyse specific locations of potential interest within an unknown environment. Concurrently, it is hypothesised that the secondary objective of the envisioned sea state estimation application, to gather additional information along the path between the goals, could also be achieved. This would lead to an enhanced understanding of the mission environment, allowing the identification of regions that require further examination, without the need for an agent to cover every possible point within the environment.

Practically, the goal locations would represent areas which are predetermined to be of interest, thus requiring an agent to further investigate that region. The agents would, moreover, gather information along the trajectories between the goals to establish if additional areas of interest exist within the environment. It is hypothesised that the development of a modified autonomous shepherding method, using HSFC paths, will improve the speed at which a set of goal locations can be reached within a fixed space. Furthermore, it is hypothesised that the introduction of the HSFC paths will permit greater coverage of the environment within a reduced time period. A comparison with the classic sheepdog shepherding technique is, therefore, conducted as a baseline for the performance of the HSFC-based method.

The main contribution of this work is an enhanced temporal efficiency for environment exploration, which is permitted when the HSFC for local environment covering is introduced to a herding task. However, the chapter also presents a unique force modulation technique to better represent the interactions between sheep and sheepdogs, as well as a circular path planning method which helps to reduce unwanted interactions between the shepherd and swarm agents.

It should be noted that the traditional shepherding task relates more specifically to a searching exercise than an exploration exercise, while the newly created technique for this study extrapolates the original task to one closer to environment exploration.

The technique developed, however, is generic and could be applied to a plethora of exploration applications, including search and rescue operations, mineral prospecting, foraging, and the scientific investigation of hostile environments.

2.2 Background Research

Swarm intelligence originates from the local interactions between a group of relatively simple agents which lead to complex behaviour of the system as a whole. The intelligent behaviours arise without any external control or regulation, defined as being an emergent property of the interactions [20]. Swarm robotic systems introduce numerous advantages, including relative simplicity in construction and programming, high robustness, and ease of scalability.

One of the first swarm control techniques was introduced by Reynolds [11] in 1987. Reynolds' control framework was based on the flocking behaviour of birds, and was originally developed to graphically simulate such behaviour. Three main behaviours were implemented to replicate the flock motion: velocity alignment, flock centring, and collision avoidance. A series of force vectors were used to enforce the three behaviours. The results closely replicated the flocking motion of real birds. Reynolds' major contribution from the research was the concept of a *boid* agent; an object which is able to sense both location and orientation information.

An ocean-based exploration study was conducted by Varughese et al. [17], who created a model which employs swarms of robotic agents to explore an underwater environment. Two different types of robotic swarms were used to conduct the exploration task: one which identifies areas of interest (the aFish), and another which stores information about the environment and identifies new regions to explore (the aMussels). The aFish mimic the scouting behaviour of bees, moving about at random until they find a source of pollen. During their simulations, if an aFish found a source of interest along the sea floor, they would scout for an aMussel, transmit the coordinates of that location to it, then the aMussel would transmit back to the aFish whether it should return to the source for further investigation, or continue scouting in a different region. The results of the authors' simulations showed that their model was effective at exploring regions of up to a hectare in size over a time period of 10–14 h. Varughese et al. noted, however, that there exists little previous research for comparison with their study.

A further bio-inspired control technique based on the herding of sheep by sheepdogs to an objective location, swarm shepherding, has also been applied to swarm agents. Strömbom et al. [16] developed a heuristic for replicating the herding behaviour of both sheep and sheepdogs which implemented the boid model and force vector-induced behaviour of Reynolds [11]. The sheep were controlled using three behaviours: sheep-to-sheep repulsion, sheep-to-sheepdog repulsion, and sheep attraction to their local centres of mass. The sheepdog enforced these behaviours on the sheep by taking one of two actions. If the sheep herd was dense enough, then the sheepdog would move into a driving position behind the herd relative to their objective destination, repelling the herd in that direction. Otherwise, the sheepdog would collect the furthest outlying sheep by moving to a position behind it relative to the herd's global centre of mass (GCM). Strömbom et al.'s simulated results showed sheep and sheepdog movements which were similar to those observed during real herding exercises.

Shepherding has now been generalised to be defined as the external guidance of a swarm, or group of agents, in order to achieve a shepherd's objective [4]. As such, shepherding has been used to control a variety of swarm systems. Of particular interest is a study by Chaimowicz and Kumar [2] who conducted a number of experiments using shepherding to command a swarm of uninhabited ground vehicles (UGVs) using multiple aerial shepherds (uninhabited aerial vehicles). The objective of the authors' research was the exploration of unknown environments, particularly for defence applications. In order to manoeuvre around obstacles and through suburban terrains, the swarm exhibited two behaviours: merging and splitting. The UGVs would split into smaller groups to fit through narrow passages and building cavities, then merge again once their surrounding environment expanded. A near-three-dimensional awareness of the environment was permitted by combining video footage from both the aerial shepherds and UGVs. While Chaimowicz and Kumar's experiments provided support for the use of shepherding as a control mechanism for swarm exploration, their methodologies and discussion lacked the depth required for replication and validation.

Furthermore, Clayton and Abbass [3] argued that all previously developed methodologies used to replicate shepherding behaviours oversimplify the complex relationship between a sheepdog and herd of sheep. As such, no standard shepherding control framework exists, and each application requires a unique shepherding control design. This research attempts to create a generalised technique for shepherd swarm guidance for the purpose of exploration.

2.3 Methodology

This section describes the simulation setup which involves a goal-based exploration task. A shepherd guides a swarm of agents to a series of goal locations within the environment. The task is repeated using a swarm act based on classical shepherding behaviours (as defined by Strömbom et al. [16]), as well as a modified approach (outlined in this section), which incorporates a localised HSFC covering behaviour for varying local environment sizes. A modulated force is introduced to better represent a sheep's response to an approaching sheepdog, as well as a simple path planning method to reduce unwanted interactions between the shepherd and swarm agents.

In each simulation, the objective of the shepherd (β) was to guide the swarm agents (π_i) to a series of goals (G_h) generated with a fixed sequential order at random within an environment. Once one of the agents reached a distance of R_G (or less) from any goal, then the goal vanishes. The shepherd achieved its objective once all goal locations disappeared. The objective of the swarming exercise, however, was only to reach every goal location within the environment.

The swarm agents were depicted using a variation of Reynolds' boids [11] and Strömbom et al.'s particle model [16]. Each swarm member was represented by an object containing an index (i), position coordinates ($P^t_{\pi_i}$), and an agent role as attributes. The shepherd agent was only characterised by its position (P^t_β). A total of N_π agents comprised the swarm, with one shepherd guiding them.

The relative locations and statuses of the swarm agents dictate the behaviour exhibited by the swarm agents themselves, as well as the behaviour of the shepherd, at each time step (t). The shepherd was able to sense the location of all of agents at time t, as well as being able to detect and dictate their roles. The swarm agents were also able to sense the location of other agents and their roles; however, they had no information regarding the objective of their motion and were not capable of decision-making.

The three main behaviours applied to the sheep in the simulations of Strömbom et al. [16] were adopted for the swarm agents in this study. Thereby, each swarm agent was repelled from each other swarm member ($\pi_{k \neq i}$) by force vector $F^t_{\pi_i \pi_k}$ if they come within a distance of $R_{\pi\pi}$ from one another (note that $R_{\pi\pi}$ represents the radius of influence of the swarm agents, not their sensing range). The swarm members were, likewise, repulsed by the shepherd when within a distance of $R_{\pi\beta}$ from them by force vector $F^t_{\pi_i \beta}$. Further, the swarm agents were attracted to their global centre of mass ($\Gamma^t_{\pi_i}$), when within sensing range ($R_{\pi\Gamma}$) of other agents, by force vector $F^t_{\pi_i \Gamma^t_{\pi_i}}$ (thus, slightly modified from [16]). The weights of the forces, $W_{\pi\pi}$, $W_{\pi\beta}$, and $W_{\pi\Gamma}$ respectively, where $W = |F|$, were calculated based on their distance from the source of the force (see Sect. 2.3.2). The overall force acting on the swarm agents when being herded was then calculated using Eq. (2.1), where $F^t_{\pi_i e}$ is a random jittering noise force which replicates the non-precise movement of sheep or robotic systems.

$$F^t_{\pi_i} = \sum_{\pi=1, k \neq i}^{\pi=N} F^t_{\pi_i \pi_k} + F^t_{\pi_i \beta} + F^t_{\pi_i \Gamma^t_{\pi_i}} + F^t_{\pi_i e} \tag{2.1}$$

Similarly, the two main behaviours of the sheepdog in Strömbom et al.'s heuristic, driving and collecting, governed the shepherd during this study. If one (or more) of the swarm agents moved outside of swarm density radius $R_{\pi\rho}$, with $\Gamma^t_{\pi_i}$ at its origin, then the shepherd moved to the target position $P^t_{T_C}$, located at distance $\mathbb{D}_Q$ behind the furthest outlying agent (π_O) with respect to $\Gamma^t_{\pi_i}$. If all agents lay within $R_{\pi\rho}$, then the shepherd moved to the target position $P^t_{T_D}$ at distance $\mathbb{D}_Q$ behind $\Gamma^t_{\pi_i}$ with respect to the current goal location. $R_{\pi\rho}$ was calculated as a function of N_π using Eq. (2.2), the same as used in [16].

$$R_{\pi\rho} = R_{\pi\pi} N^{2/3} \tag{2.2}$$

The novel aspect of this study, departing from that of Strömbom et al. [16], was to assign each swarm agent one of four roles: *herd member*, *immobilised*, *torpedo*, or *explorer*. During an exploration exercise, the herd members navigated their environment according to the aforementioned three classical shepherding behaviours, being influenced by the repulsion and attraction forces of the swarm and shepherd (the only role present in [16]). However, if an agent reached a goal location (within distance R_G), which was not required to be the current goal sought by the shepherd, then the agent's role would switch to immobilised, and it would become stuck at the goal location for a count of T_f time steps. Once the T_f steps are complete, the agent's role switches to torpedo. The agent is then pushed by force vector $F^t_{\pi_i B}$, of constant weight $W_{\pi B}$, in a randomly generated direction for a count of T_b time steps, as a means for further environment coverage.

Following the T_b steps, the agent's role will either be converted back to a herd member, or switches to an explorer (depending on its distance from $\Gamma^t_{\pi_i}$). The final role, explorer, is initialised if an agent reaches a radial distance of R_H from $\Gamma^t_{\pi_i}$. At this point, it is assumed that the agent is outside of sensing range of the other swarm agents, and the shepherd recognises it as separate from the rest of the swarm, which it is collecting and driving to the next goal location. Therefore, any swarm agents outside of R_H are not included in the calculation of $\Gamma^t_{\pi_i}$. The final role, explorer, is initialised if an agent reaches a radial distance of R_H from $\Gamma^t_{\pi_i}$. Once assigned the explorer role, the agent restricts itself to a local environment of size L_l and begins following a Hilbert space-filling curve (HSFC), of order H_O (more information on HSFCs can be found in Sect. 2.3.4), acting as an area covering path. Once the path is complete, the agent's role will switch to a torpedo. The swarm role assignment methodology is captured in Algorithm 1, where roles are assigned as herd member (1), immobilised (2), torpedo (3), and explorer (4), updated at each time step.

Figure 2.1 gives an example of a simulation, where the shepherd and three swarm agents are shown with their corresponding repulsion force radii (and their influential spheres shown surrounding them). The outlying agent (π_O) is outside of R_H, therefore, its role would switch to explorer, while the rest of the agents are herd members, lie within $R_{\pi\rho}$, and are, therefore, dense enough to be driven towards the goal.

The swarm agents were randomly initialised in the north-east quadrant of the environment at positions $P^{t=1}_{\pi_i}$, and the shepherd was initialised in the south-west quadrant at position $P^{t=1}_{\beta}$. All swarm members began with the role of a herd member.

2.3.1 Simulation Setup

Two types of shepherding were compared during the simulations. The first was a general sheepdog herding sheep to multiple goals shepherding exercise, where only a single sheep was required to reach any single goal location, rather than the entire herd. The second introduced a new method for exploring an environment using goal locations and HSFC paths.

The second set of simulations, which introduce the Hilbert exploration technique, have the same objective as the first, to herd the swarm agents to each goal location.

Algorithm 1 Swarm agent role assignment

for all i **do**
  for all h **do**
    if $\| P^t_{\pi_i} - P_{G_h} \| \leq R_G$ **then**
      $\pi_i(role) == 2$
      $\pi_i(immobilise\_count) = 0$
    end if
  end for
  if $\pi_i(role) == 1$ **then**
    if $\| P^t_{\pi_i} - \Gamma^t_{\pi_i} \| \geq R_H$ **then**
      $\pi_i(role) == 4$
      $\pi_i(HSFC\_position) = 0$
    end if
  else if $\pi_i(role) == 2$ **then**
    if $\pi_i(immobilise\_count) == T_f$ **then**
      $\pi_i(role) == 3$
      $\pi_i(torpedo\_count) = 0$
    else
      $\pi_i(immobilise\_count) = \pi_i(immobilise\_count) + 1$
    end if
  else if $\pi_i(role) == 3$ **then**
    if $\pi_i(torpedo\_count) == T_b$ **then**
      if $\| P^t_{\pi_i} - \Gamma^t_{\pi_i} \| \geq R_H$ **then**
        $\pi_i(role) == 4$
        $\pi_i(HSFC\_position) = 0$
      else
        $\pi_i(role) == 1$
      end if
    else
      $\pi_i(torpedo\_count) = \pi_i(torpedo\_count) + 1$
    end if
  else if $\pi_i(role) == 4$ **then**
    if $\pi_i(HSFC\_position) == (length(HSFC\_path) - 1)$ **then**
      $\pi_i(role) = 3$
      $\pi_i(torpedo\_count) = 0$
    else
      $\pi_i(HSFC\_position) = \pi_i(HSFC\_position) + 1$
    end if
  end if
end for

However, in this setup, if an agent strays a distance of $R_{\pi H}$ from $\Gamma^t_{\pi_i}$, then it re-initialises itself within a local environment and starts following an HSFC path, hoping to reach one of the goal locations along its trajectory.

If the swarm agent reaches one of the goals, then it becomes bound to the goal (as the straight herding setup), before being torpedoed away. Similarly, if an agent arrives at the end of a HSFC path without encountering any goals, then the agent gets torpedoed in a random direction and either rejoins the herd or begins following a new HSFC path.

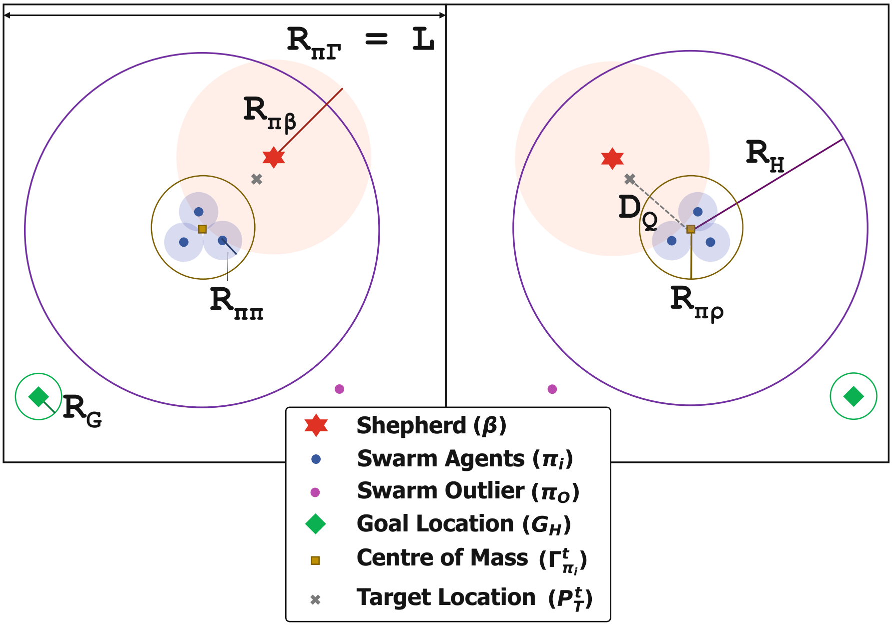

Fig. 2.1 Split mirror illustrations of experimental setup including a shepherd, swarm agents, and a goal. The left image, additionally, shows the shepherd repulsion radius, sheep repulsion radius, goal radius, and environment length and global centre of mass attraction radius. The right image includes the Hilbert radius, swarm density radius, and shepherd target distance. Note that the parameters are not correctly scaled

Table 2.1 outlines each of the simulation parameters and their corresponding values used. As this study is the first of its kind, the environment size, swarm repulsion radius, shepherd repulsion radius, swarm density radius, and shepherd and sheep step size were all given the same values as used by Strömbom et al. [16] as a foundation, while the other values were chosen intuitively to complement Strömbom et al.'s original study.

The first experiment undertaken was to test the effect of the size of the localised HSFC environment (L_l) on the time it took to reach 10 goals for varying maximum influential force weight (W_{max}). Three L_l values were simulated: $L_l = \frac{L}{N}$, $L_l = 0.50 * \frac{L}{N}$, and $L_l = 1.50 * \frac{L}{N}$. Force weightings between $10N$ and $200N$ were tested at intervals of $10N$ separately for $W_{\pi\pi}$, $W_{\pi\beta}$, and $W_{\pi\Gamma}$. While one maximum weighting was modified, the other two were fixed at $100N$.

The second experiment was completed to examine how the number of goals (N_G) within a fixed environment, and thus goal density, affected the performance of the developed HSFC exploration technique. N_G values ranging from 5 to 50 were simulated at intervals of 5 for each of the local environment lengths described above.

While the goals were generated with the perimeter of the environment, the swarm agents and shepherd were not restricted to boundaries. Each simulation setup was

Table 2.1 Parameters for shepherding experiments

Parameter	Value
Maximum number of time steps (t_{max})	5000
Environment length (L)	150m
Number of swarm agents (N_π)	10
Swarm repulsion radii ($R_{\pi\pi}$)	2m
Swarm speed (V_π)	$1\frac{m}{t}$
Swarm density radius ($R_{\pi\rho}$)	9.28m
Number of shepherds (N_β)	1
Shepherd repulsion radius ($R_{\pi\beta}$)	65m
Shepherd speed (V_β)	$1.5\frac{m}{t}$
Shepherd target distance ($\mathbb{D}_Q$)	40m
Shepherd alignment distance ($\mathbb{D}_A$)	25m
GCM attraction radii ($R_{\pi\Gamma}$)	150m
Noise force weight ($W_{\pi_i n}$)	0.3N
Number of goals (N_G)	10
Goal radius (R_G)	5m
Circular path radius (R_{CO})	65m
Circular path angular speed ($\Delta\theta_\beta$)	$0.025 rad/t$
Fixed count (T_f)	100t
Torpedo force weight ($W_{\pi B}$)	100N
Torpedo force count (T_b)	20N
HSFC order (H_O)	3
Hilbert radius (R_H)	39.28m

repeated 30 times with different seeds, then the results for that setup were averaged. The simulations were run until all goals had been reached, or the maximum of 5000 time steps was completed. Both experiments were compared to the performance of the pure herding shepherding to reach all goals. Both experiments compare the HSFC-based shepherding to the classic shepherding (derived from [16]).

2.3.2 *Force Modulation*

Strömbom et al. [16] applied a discrete force when a sheep fell within the ranges of influence of other agents. In this study, it was assumed that the influence force weightings acting on a swarm agent were applied with a natural exponential trend. The repulsion force weakens, and attraction force strengthens, as the radial distance from their source increased. Equation (2.3) defines the repulsion force as a function of radial distance ($r_{\pi R}$) from the repulsion force origin, while Eq. (2.4) describes the attraction force as a function of radial distance ($r_{\pi A}$) from the attraction source. $W_{\pi R}$ is the modulated weight of the repulsion force and W_{Rmax} is the maximum

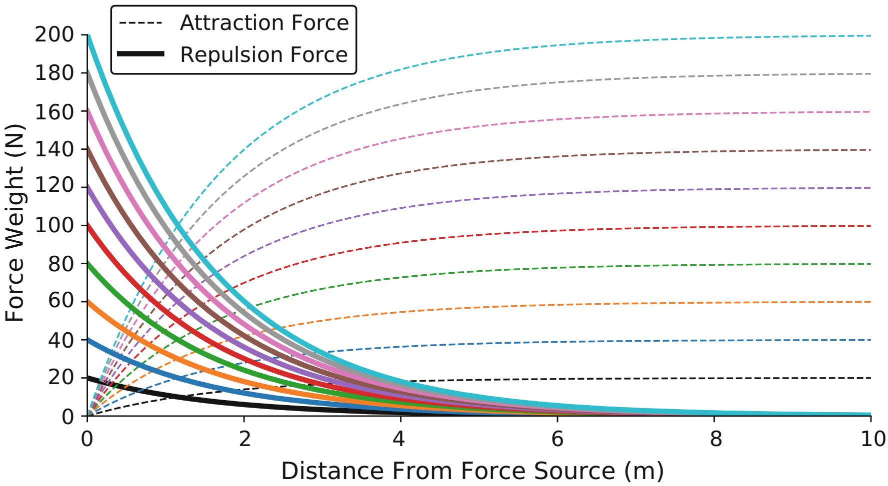

Fig. 2.2 Influential force modulation for a series of maximum force weights. Each colour represents the matching repulsive and attractive force weights applied at a given radius

force weight (replaced by $W_{\pi\pi}$ or $W_{\pi\beta}$). Likewise, $W_{\pi A}$ is the modulated weight of the attraction force and $W_{\pi\Gamma}$ is the maximum possible attraction force strength. $R_{\pi R}$ represents the radii $R_{\pi\pi}$ and $R_{\pi\beta}$. Figure 2.2 depicts the natural exponential relationship between force strength and radial distance from the source (here, at the origin [0,0]), for an influential radius of $6m$. Beyond $6m$, the force tends to $0N$, so has no significant effect on the behaviour of the swarm agents.

$$W_{\pi R} = W_{Rmax} * e^{\left(\frac{-r_{\pi R}}{R_{\pi R}} * 6\right)} \tag{2.3}$$

$$W_{\pi A} = W_{\pi\Gamma} * \left(1 - e^{\left(\frac{-r_{\pi\Gamma}}{R_{\pi\Gamma}} * 6\right)}\right) \tag{2.4}$$

2.3.3 Path Planning

As discussed, when herding sheep, a sheepdog exhibits two main behaviours: driving and collecting [16]. When the shepherd moves to a target location P_T^t (representing $P_{T_C}^t$ or $P_{T_D}^t$), it must position itself behind the swarm with respect to the direction of G_h, or behind π_O with respect to $\Gamma_{\pi_i}^t$. As such, in order to achieve this without driving the swarm agents towards an undesired direction, the shepherd must plan its path to P_T^t.

A simple method of achieving this was to create a circular path with its origin (P_{CO}^t) centred on either the $\Gamma_{\pi_i}^t$ or π_O, with a radius large enough to render the influential force of the shepherd ineffective (in this case, $R_{\pi\beta}$). With each step, the shepherd calculates its distance $\mathbb{D}_T$ from P_T^t, and distance $\mathbb{D}_{CO}$ from P_{CO}^t. The shepherd moves towards P_T^t until it reaches a point along the circular path, within a tolerance of ϵ, such that $P_{CO}^t = R_{\pi\beta} \pm \epsilon$. The shepherd then follows the path until positioned behind its desired P_T^t, finally moving towards it once within an alignment distance of $\mathbb{D}_A$ from it, where $\mathbb{D}_A = R_{\pi\beta} - \mathbb{D}_Q$. If the shepherd already lies within the circle, then it simply moves in a straight path towards P_T^t. Algorithm 2 gives a breakdown of the path planning algorithm used.

Algorithm 2 Circular-path planning algorithm

while $t < 5000$ **do**
  Calculate $\mathbb{D}_T$
  Calculate $\mathbb{D}_{CO}$
  if $\mathbb{D}_T < \mathbb{D}_A$ **then**
    $P_\beta^{t+1}(x, y) = P_\beta^t(x, y) + V_\beta * \frac{P_T^t(x,y) - P_\beta^t(x,y)}{\| P_T^t(x,y) - P_\beta^t(x,y) \|}$
  else if $\mathbb{D}_T > (R_{\pi\beta} + \epsilon)$ **then**
    $P_\beta^{t+1}(x, y) = P_\beta^t(x, y) + V_\beta * \frac{P_T^t(x,y) - P_\beta^t(x,y)}{\| P_T^t(x,y) - P_\beta^t(x,y) \|}$
  else
    $P_\beta^{t+1}(x) = P_{CO}^t + R_{\pi\beta} * \cos \Delta\theta_\beta$
    $P_\beta^{t+1}(y) = P_{CO}^t + R_{\pi\beta} * \sin \Delta\theta_\beta$
  end if
end while

Figure 2.3 gives an example simulation of such a path. The simulation was such that the swarm, of size $N_\pi = 5$, was forced to remain stationary to prove the path planning concept. Once initialised (where $P_\beta^{t=1}$ is represented by a larger marker), the shepherd makes a decision on whether to drive the herd towards the single goal at $P_G = [10, 10]$, or collect an outlier agent. In this case, the swarm is dense enough to drive, so the shepherd approaches $P_{T_D}^t$ until it reaches the radial distance $R_{\pi\beta}$ from $\Gamma_{\pi_i}^t$, then tracks along the circular path (represented by the thin black line) until it arrives at a location approximately behind $P_{T_D}^t$ and moves towards it. $R_{\pi\beta}$ was set to $30m$ and $\mathbb{D}_A$ was set to $10m$. R_G is displayed as the green circle surrounding the goal.

2.3.4 *Hilbert Space-Filling Curves*

A space-filling curve, within a square boundary, is a continuous path which moves through each point within the square. Concepts for space-filling curves were originally invented by Peano [10], Hilbert [7], and Sierpinski [12].

Hilbert curves are one of the simplest space-filling, or fractal, curves as there are only four directions that the path can move in at any given step [5]. These directions can be represented as the four cardinal directions: north (Nth), south (Sth), east (Est), and west (Wst). The shapes constructed during the Hilbert space-filling walk

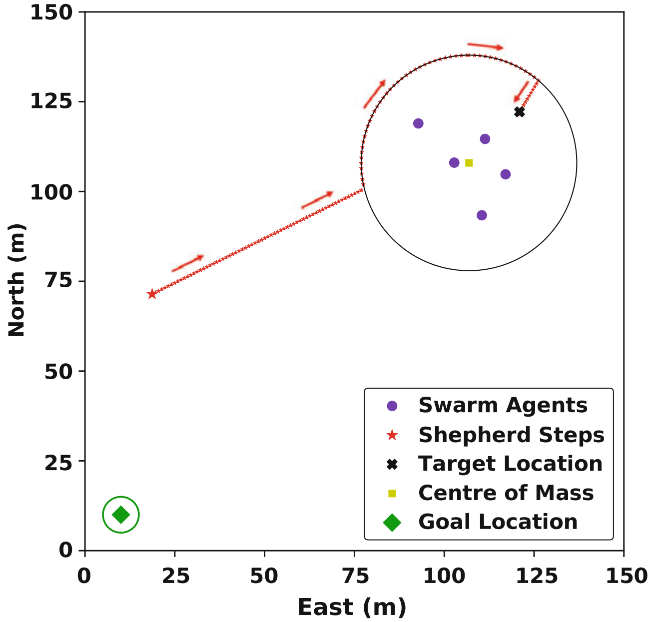

Fig. 2.3 Shepherd circular path to driving target location

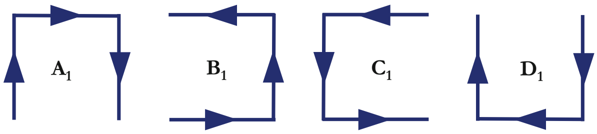

Fig. 2.4 First order Hilbert space-filling curve path directions

are repeated square patterns where one edge is removed. The path is followed in a clockwise direction if the square has only one horizontal edge, and in an anti-clockwise direction if it has only one vertical edge. The four open-square patterns which are formed along the walk, collectively defined as the first order HSFC (H_1), and their corresponding directions of motion, are given in Fig. 2.4.

These units are represented by the identifiers: A_1, B_1, C_1, and D_1. For a second order Hilbert curve (H_2), two of the H_1 units are linked together, and joined to other H_1 units which are traversed in an opposite sense. Wirth [19], then, defined the set of Eqs. (2.5)–(2.8) to find an mth order HSFC, where each sequential order is comprised of sets of the previous order with alternating senses. The first to fourth order HSFCs are shown in Fig. 2.5.

$$A_{m+1} = [B_m, Nth, A_m, Est, A_m, Sth, C_m] \tag{2.5}$$

$$B_{m+1} = [A_m, Est, B_m, Nth, B_m, Wst, D_m] \tag{2.6}$$

$$C_{m+1} = [D_m, Wst, C_m, Sth, C_m, Est, A_m] \tag{2.7}$$

$$D_{m+1} = [C_m, Sth, D_m, Wst, D_m, Nth, B_m] \tag{2.8}$$

The HSFC was selected as the covering method for this study due to its simplicity, which matches the advantages of using a swarm-based system for exploration, where each swarm agent should remain rudimentary in its own capacity to maintain the robustness and flexibility of swarm system as a whole. Further, the third order HSFC was used to match the L_l, L, and R_G used, with lesser orders not covering enough points within the environment, and larger orders covering too many.

2.4 Results and Discussion

A preliminary simulation was run for both the classic herding (Fig. 2.6) and Hilbert shepherding (Fig. 2.7) setups in order to give an example comparison of the methods. The same goals were generated, which followed the same sequence

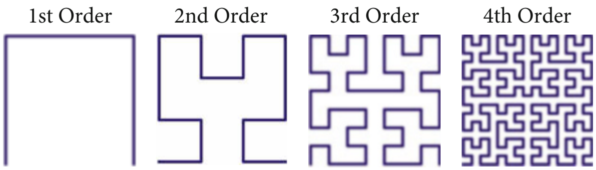

Fig. 2.5 Hilbert space-filling curves of orders 1–4

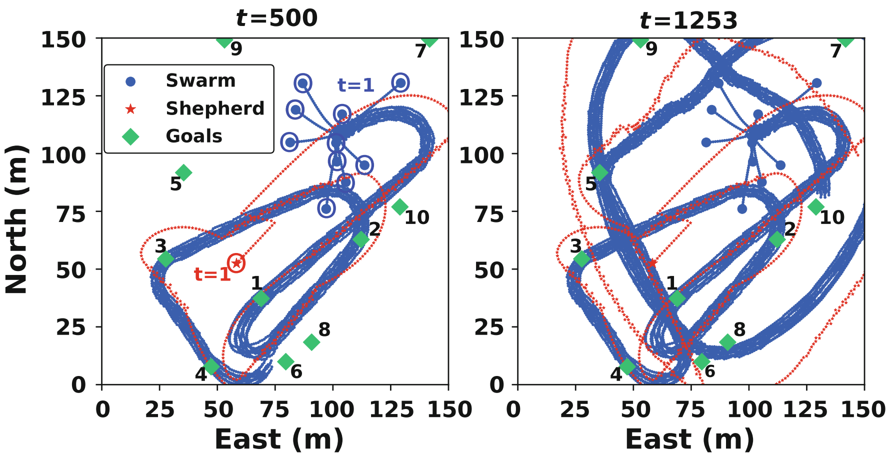

Fig. 2.6 Herding swarm shepherding exploration paths for $t = 500$ and $t = 1253$ time steps, and with goals G_1 to G_{10}

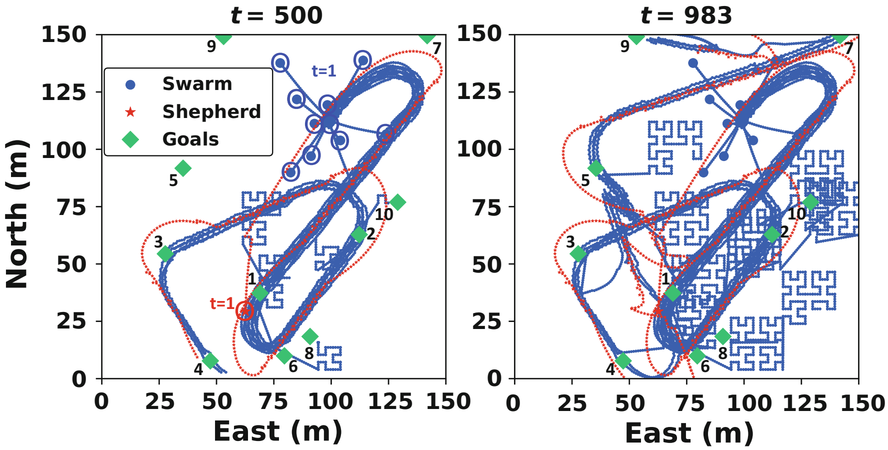

Fig. 2.7 HSFC-based swarm shepherding exploration paths for $t = 500$ and $t = 983$ time steps, and with goals G_1 to G_{10}

Table 2.2 Goal coordinates and order for example simulations

Goal (G_H)	Eastern coordinate (m)	Northern coordinate (m)
1	68.96	37.28
2	112.22	62.81
3	27.69	54.42
4	47.31	7.75
5	35.43	91.74
6	79.62	9.94
7	141.84	149.57
8	90.80	18.37
9	53.10	149.23
10	128.99	76.92

(given in Table 2.2), and the total paths of the shepherd and swarm agents were plotted. Finally, the maximum influential force weights were equated as $W_{\pi\pi} = W_{\pi\beta} = W_{\pi\Gamma} = 100N$.

The pure herding case, given in Fig. 2.6, is split into two sections. The first shows the overall paths for the shepherd and swarm agents at $t = 500$, and the second displays the completion of the exercise, at $t = 1253$. $P_{\pi_i}^{t=1}$ and $P_{\beta}^{t=1}$ are represented by a larger marker than was used for their corresponding paths, with a surrounding circle included on the left image, while each of the goals is numbered (as given in Table 2.2). The motion of the shepherd was found to be quite similar to that of Strömbom et al. [16] when driving and collecting the agents, with a small-scale zigzag type manoeuvre observed keeping the herd centred about $\Gamma_{\pi_i}^t$. Upon initialisation, the swarm agents were quickly attracted to $\Gamma_{\pi_i}^t$, forming a herd. The shepherd moved towards a target position at first (likely $P_{T_C}^t$), before briefly following the circumference of the circular path, then proceeded to move towards $P^t_{T_D}$ as the herd condensed and it fell within $R_{\pi\beta}$. The agents were then herded to G_1 to G_4 in the prescribed order, before reaching G_6 at $\sim t = 500$. In the final graph, it can be seen that the shepherd herds the swarm to the goals in the correct sequential order (once G_6 was disregarded as it had already been reached), passing outside of the environment bounds briefly between G_7 and G_8, then again between G_9 and G_{10} (due to the goals being generated near the borders). The shepherd spent very little time following the circumference of the circle defined by the path planning algorithm, though, as once it reached a P^t_T, it would generally remain within a distance of $R_{\pi\beta}$ from the circular path origin, instead continually moving along a straight line towards P^t_T. The reduction of $R_{\pi\beta}$ could solve this problem and further mitigate unwanted repulsion by the shepherd.

An equivalent simulation for the HSFC-based exploration technique is given in Fig. 2.7 for $L_l = 22.5$ m at $t = 500$ and $t = 983$ (the end of the simulation). At $t = 500$ it can be seen that the path of the herding agents begins quite similar to that of the pure herding case, with the herd of agents reaching G_1 to G_4. However, the agent which became immobilised at G_2 had torpedoed north-east, switched to an explorer, then quickly reached G_{10}. Similarly, an agent which became immobilised at G_6 was torpedoed south-east before following an HSFC path until it reached G_8. A second agent, after becoming immobilised at G_1 for time T_f, had begun following a Hilbert path, while two other agents had become immobilised at G_1. At the completion of the environment exploration, it can be seen that the agents which reached G_6 and G_{10} remained outside of R_H stayed within the same regions and continued following HSFC paths. Likewise, the agents which were torpedoed from G_4 and G_5 started exploring without rejoining the herd, while the agent immobilised at G_3 was collected before G_7 and G_9 were finally covered. As with the herding case, there was only a very brief moment, just after the shepherd was initialised, when it followed the circumference of the circular path.

It can be seen that the Hilbert exploration setup was able to reach the 10 goals faster than the herding exploration setup, taking 78% of the number of time steps. It also appears that the HSFC-based method covered more of the environment over the shorter time period. While no reliable conclusions can be made from one simulation, the example gives a good indication of the pattern which emerged from the extensive experimental campaign.

2.4.1 Force Weights

Figure 2.8 displays the amount of time it took to reach 10 goals for the pure herding and Hilbert shepherding exploration experiments, averaged over the 30 simulations run for each of the HSFC local environment sizes and classic herding cases. This was done for the three local environment sizes, over each of the force weight ranges tested for $W_{\pi\pi}$, $W_{\pi\beta}$, and $W_{\pi\Gamma}$. The size of L_l, however, had little affect on the time taken to cover all of the goals for any of the maximum force weights.

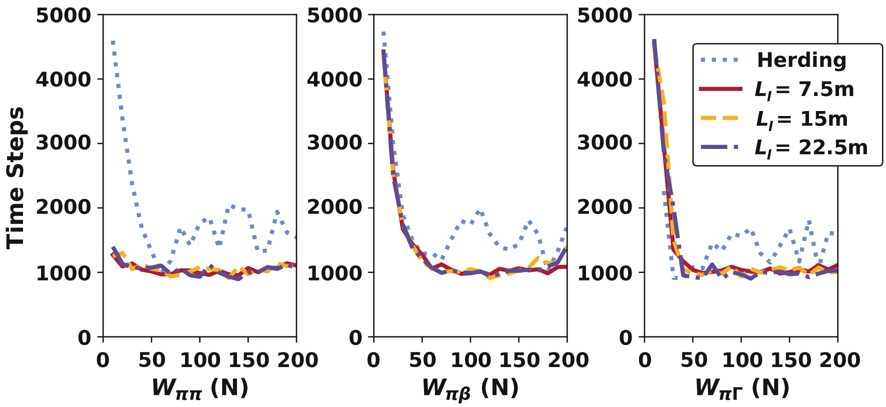

Fig. 2.8 Maximum force weighting versus time

When varying $W_{\pi\pi}$, the HSFC-based exploration technique outperformed the pure herding method each for all weights, except at $W_{\pi\pi} = 60$, where the herding method was able to reach all 10 goals slightly quicker than the Hilbert method for $L_l = 22.5m$. The Hilbert technique was able to cover all goals much faster than the traditional herding method for low sheep-to-sheep repulsion values, likely due to the herd forming a much tighter group, thus, requiring more precise guidance to attain a goal. The performance of each of the methods remained the same for low $W_{\pi\beta}$ values, as the herd would have been less likely to break apart due to the presence of the shepherd. As the weight increased, the Hilbert method was shown to be more efficient. When the maximum weight for attraction to $\Gamma^t_{\pi_i}$ was $10N$, the shepherd was unable to guide the swarm to all 10 goals. This showcases the difficulty confronting the shepherd if the swarm agents are not attracted to their herd, where collection would likely become more time consuming and take up the majority of the shepherd's energy. In general, the maximum repulsion and attraction force weightings were found to not significantly affect the temporal efficiency of the HSFC-based shepherding task beyond a value of approximately $50N$; however, the classic shepherding case was found to fluctuate significantly, likely due to the greater percentage of swarm agents acting as herd members.

2.4.2 Number of Goals

The effectiveness of the developed HSFC exploration method increased as the number of goals in the environment increased. This can be explained by the fact that the HSFC-based technique results in more coverage of an environment over a shorter time period (as can be seen in Figs. 2.7 and 2.8). Therefore, with a higher density, the Hilbert paths became more advantageous. The traditional herding exploration method was unable to cover 50 goals within the 5000 time steps, and barely able

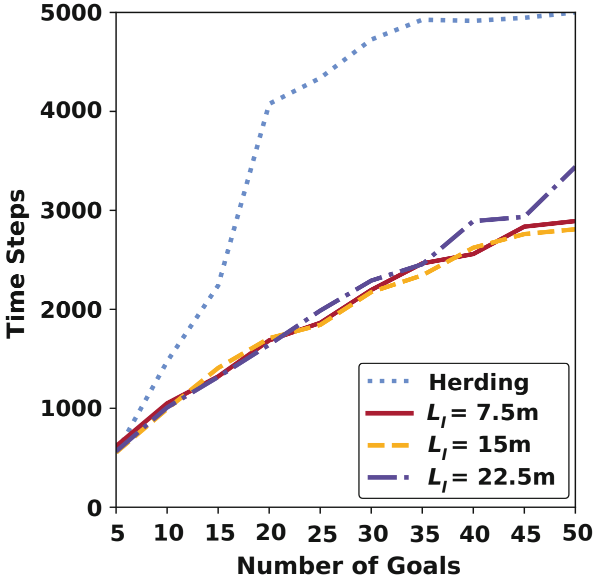

Fig. 2.9 Number of goals versus time

to reach more than 35 goals. The performance of each of the L_l values remained fairly constant until N_G increased above 35, at which point the larger Hilbert covering area ($L_l = 22.5m$) reduced the efficiency of the task. If there are only a few goal destinations required to be visited within an area (for $N_G = 5$), then the herding method performed just as well as the Hilbert method. The quantity of areas of interest within a given environment should, therefore, dictate the type of exploration technique used. If there are a few very specific areas of significance requiring exploration within an environment, then classic shepherd herding control should be adopted, however, for regions with more uncertainty or a vast quantity of goal destinations, then the Hilbert method was shown to be more appropriate. The local environment size for the HSFCs must then be considered, where one too large may not perform as well given a very high goal density (Fig. 2.9).

2.5 Conclusion

The use of shepherding for environment exploration using a goal-based system has been investigated. The autonomous swarm shepherding method developed using HSFC paths appears to be more effective at exploring an environment than traditional herding using goal-based waypoints. The time taken to cover all goals within the mission environment was significantly lower for the majority of maximum force weights tested. The improvements upon the herding technique became more apparent with the greater number of goals within the environment. Further, it seemed as though the Hilbert method enabled greater coverage of the environment than the traditional herding technique.

The simple circular path planning algorithm created for the exploration task to enable the shepherd to manoeuvre around the swarm agents without imparting unwanted force on them appeared to have been used primarily at the beginning of the exercise, then seldom again. More research must be conducted to investigate the effect of the circular path radius as the shepherd often lay within the circular path during the simulations, signifying that it rarely tracked along its circumference.

Future work will test the effects of the HSFC order on the performance of the goal-based exploration task, as well as the variation of the number of swarm agents involved. Moving beyond the described setup, additional research should address dynamic goals, uncertainty in their locations and seek to compare single and multi-agent approaches for exploration to single and multi-shepherding agent approaches.

The initial results, however, show promise for the use of shepherding guidance of swarms as a solution for autonomous environment exploration. Furthermore, the presented simulations have given an example of the feasibility of the sea state determination problem.

References

1. Burgard, W., Moors, M., Fox, D., Simmons, R., Thrun, S.: Collaborative multi-robot exploration. In: Proceedings 2000 ICRA. Millennium Conference. IEEE International Conference on Robotics and Automation. Symposia Proceedings, pp. 476–481 (2000)
2. Chaimowicz, L., Kumar, V.: Aerial shepherds: Coordination among UAVS and swarms of robots. In: Alami, R., Chatila, R., Asama, H. (eds.) Distributed Autonomous Robotic Systems, vol. 6, pp. 243–252. Springer Japan, Tokyo (2007)
3. Clayton, N.R., Abbass, H.: Machine teaching in hierarchical genetic reinforcement learning: Curriculum design of reward functions for swarm shepherding. In: Proceedings of the IEEE Congress on Evolutionary Computation (2019)
4. Gee, A., Abbass, H.: Transparent machine education of neural networks for swarm shepherding using curriculum design. In: Proceedings of the International Joint Conference on Neural Networks (2019)
5. Griffiths, J.: Table-driven algorithms for generating space-filling curves. Comput.-Aided Design **17**(1), 37–41 (1985)
6. Gu, X., Moan, T.: Long-term fatigue damage of ship structures under nonlinear wave loads. Marine Technol. **39**(2), 95–104 (2002)
7. Hilbert, D.: Über die stetige Abbildung einer Linie auf ein Flächenstück. In: Dritter Band: Analysis· Grundlagen der Mathematik· Physik Verschiedenes, pp. 1–2. Springer, Berlin (1935)
8. Lien, J.M., Bayazit, O.B., Sowell, R.T., Rodriguez, S., Amato, N.M.: Shepherding behaviors. In: IEEE International Conference on Robotics and Automation, vol. 4, pp. 4159–4164. Citeseer (2004)
9. Long, N.K., Sammut, K., Sgarioto, D., Garratt, M., Abbass, H.A.: A comprehensive review of shepherding as a bio-inspired swarm-robotics guidance approach. IEEE Trans. Emer. Topics Comput. Intell. **4**, 523–537 (2020)
10. Peano, G.: Sur une courbe, qui remplit toute une aire plane. Math. Ann. **36**(1), 157–160 (1890)

11. Reynolds, C.: Flocks, herds and schools: A distributed behavioral model. In: Proceedings of the 14th Annual Conference on Computer Graphics and Interactive Techniques, SIGGRAPH ’87, pp. 25–34. ACM, New Yrok (1987)
12. Sierpínski, W.: Sur une nouvelle courbe continue qui remplit toute une aire plane. Bull. Acad. Sci. Cracovie (Sci. Math. et Nat. Serie A) 462–478 (1912)
13. Smith, M.: I Am Just Going Outside: Captain Oates-Antarctic Tragedy. Gill & Macmillan, Dublin (2002)
14. Stella, A.C., Vakkalanka, J.P., Holstege, C.P., Charlton, N.P.: The epidemiology of caving fatalities in the United States. Wild. Environ. Med. **26**(3), 436–437 (2015)
15. Stevens, S.C., Parsons, M.G.: Effects of motion at sea on crew performance: a survey. Marine Technol. **39**(1), 29–47 (2002)
16. Strömbom, D., Mann, R.P., Wilson, A.M., Hailes, S., Morton, A.J., Sumpter, D.J.T., King, A.J.: Solving the shepherding problem: heuristics for herding autonomous, interacting agents. J. R. Soc. Interf. **11**(100) (2014). https://browzine.com/articles/52614503
17. Varughese, J.C., Thenius, R., Leitgeb, P., Wotawa, F., Schmickl, T.: A model for bio-inspired underwater swarm robotic exploration. IFAC-PapersOnLine **51**(2), 385–390 (2018)
18. Whaite, P., Ferrie, F.P.: Autonomous exploration: driven by uncertainty. IEEE Trans. Pattern Anal. Mach. Intell. **19**(3), 193–205 (1997). https://doi.org/10.1109/34.584097
19. Wirth, N.: Algorithms+ data structures= programs. Technical Report (1976)
20. Zoghby, N.E., Loscri, V., Natalizio, E., Cherfaoui, V.: Chapter 8: Robot Cooperation and Swarm Intelligence, Chap. 8, pp. 163–201 (2013). https://doi.org/10.1142/9789814551342_0008